The V-tail equivalence

$S_{h,\mathrm{eff}} = \sum_i S_i \cos^2 \nu_i$ and $S_{v,\mathrm{eff}} = \sum_i S_i \sin^2 \nu_i$ in linear theory

da3Dalus calculation canon — derivation note for GH #1208

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Abstract

A V-tail panel of true (developed) area S and lift-curve slope a, inclined by the dihedral angle ν, contributes $aS\cos^2\nu$ to the pitch-force derivative and $aS\sin^2\nu$ to the side-force derivative. One factor $\cos\nu$ (or $\sin\nu$) comes from projecting the free-stream incidence onto the panel normal, the other from projecting the panel's normal force back onto the body axes. This note derives the result step by step, generalises it to polyhedral tails, lists the limit checks, and separates what is exact in linear theory from what is an assumption.

1 Step 1 — Axes and geometry

Assumptions. Rigid aircraft; body axes with x forward, y to the right (starboard), z *up*. (Standard flight-mechanics body axes have z down; that only flips the sign of the vertical force Z.) The tail is mirror-symmetric about the x–z plane. Each panel is flat, with true area S (measured in its own plane) and dihedral angle ν to the horizontal.

Geometry. The spanwise unit vectors $\mathbf{s}$ and the panel normals $\mathbf{n}$ (in (x, y, z) components) are

$$\mathbf{s}_R = (0,\ \cos\nu,\ \sin\nu), \qquad \mathbf{n}_R = (0,\ -\sin\nu,\ \cos\nu), \tag{1}$$

$$\mathbf{s}_L = (0,\ -\cos\nu,\ \sin\nu), \qquad \mathbf{n}_L = (0,\ +\sin\nu,\ \cos\nu). \tag{2}$$

$\mathbf{n}_R$ is the vertical unit vector $\mathbf{e}_z$ rotated by $+\nu$ about x; $\mathbf{n}_L$ is $\mathbf{e}_z$ rotated by $-\nu$. Both normals point "up and inboard" (Fig. 1).

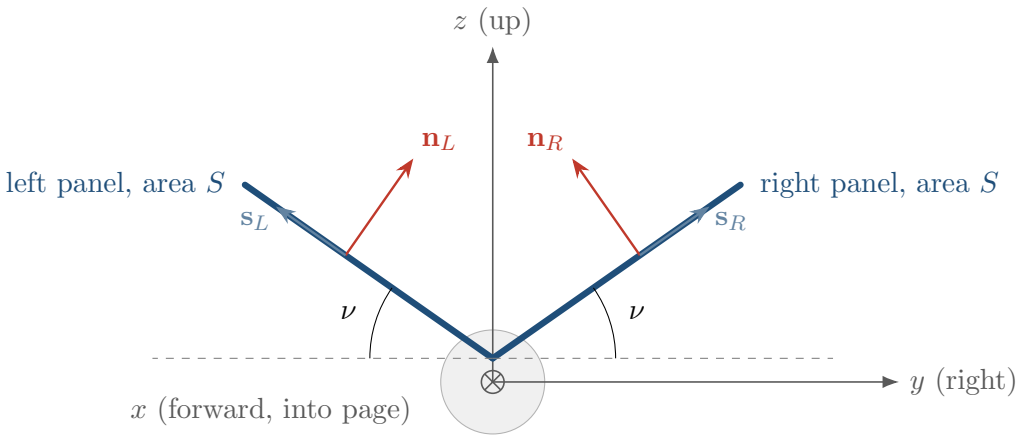


Figure 1: Rear view (looking forward). Each panel lies at dihedral ν to the horizontal; its normal $\mathbf{n}$ is the vertical axis rotated by $\pm\nu$ about x. Drawn for $\nu = 35°$.

2 Step 2 — Incidence normal to the panel

Assumptions. Small angle of attack α and small sideslip β (with $\beta > 0$ meaning wind from starboard); α and β are the *local* values at the tail, i.e. they already contain downwash and sidewash. First order in α, β throughout.

Equation. The air velocity relative to the aircraft is

$$\mathbf{V}_{\text{air}} = V\,(-\cos\alpha\cos\beta,\ -\sin\beta,\ \sin\alpha) \approx V\,(-1,\ -\beta,\ \alpha). \tag{3}$$

The chordwise component is V to first order, so the panel's normal incidence is $\alpha_n = \mathbf{V}_{\text{air}} \cdot \mathbf{n}/V$. With (1)–(2):

$$\boxed{\alpha_{n,R} = \alpha\cos\nu + \beta\sin\nu, \qquad \alpha_{n,L} = \alpha\cos\nu - \beta\sin\nu.} \tag{4}$$

In words. An upward crossflow $V\alpha$ meets a tilted panel; only its component along the normal, $V\alpha\cos\nu$, is an incidence. A crossflow from starboard, $V\beta$, has the component $V\beta\sin\nu$ along the normals, with opposite sign on the two panels: it pushes the right panel on its outer (lower) face and the left panel on its inner (upper) face (Fig. 2).

(a) angle of attack: crossflow $V\alpha$ upward (b) sideslip: crossflow $V\beta$ from starboard

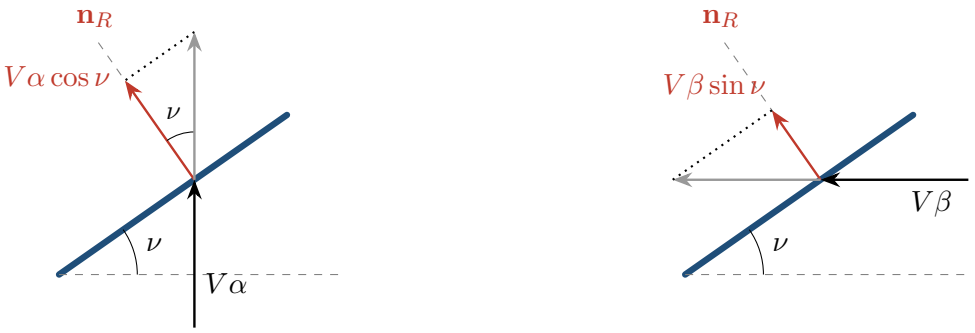


Figure 2: Projection of the crossflow onto the normal of the right panel (rear view). Grey arrow: crossflow vector moved to the panel; red: its component along $\mathbf{n}_R$. (a) $\alpha_n = \alpha\cos\nu$; (b) $\alpha_n = +\beta\sin\nu$ on the right panel, $-\beta\sin\nu$ on the left (mirror image, not drawn).

3 Step 3 — Normal force and its projection

Assumptions. Linear, attached flow. The panel force is normal to the panel: by Kutta–Joukowski $\mathbf{F} = \rho\,\mathbf{V} \times \boldsymbol{\Gamma}$ with $\mathbf{V} \parallel x$ and $\boldsymbol{\Gamma} \parallel \mathbf{s}$, so $\mathbf{F} \parallel \mathbf{n}$. The normal-force slope of the panel is a (per radian, based on the true area S); see Step 4 for which a.

Equation.

$$\mathbf{F}_i = q\,S\,a\,\alpha_{n,i}\,\mathbf{n}_i, \qquad q = \tfrac{1}{2}\rho V^2, \quad i \in \{R, L\}. \tag{5}$$

Projecting onto z and y with (1)–(2) and inserting (4):

$$Z = qSa\cos\nu\,(\alpha_{n,R} + \alpha_{n,L}) = 2qSa\,\alpha\cos^2\nu, \tag{6}$$

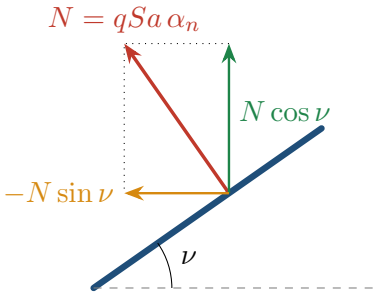
$$Y = qSa\sin\nu\,(\alpha_{n,L} - \alpha_{n,R}) = -2qSa\,\beta\sin^2\nu. \tag{7}$$

With $S_{\text{tot}} = 2S$:

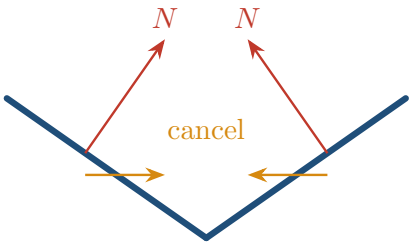
$$\boxed{\frac{\partial Z}{\partial\alpha} = q\,a\,S_{\text{tot}}\cos^2\nu, \qquad \frac{\partial Y}{\partial\beta} = -q\,a\,S_{\text{tot}}\sin^2\nu.} \tag{8}$$

In words. The first $\cos\nu$ is the incidence projection (4); the second is the projection of the normal force onto the vertical (Fig. 3). The same holds with $\sin\nu$ for sideslip. The cross terms cancel for the mirror pair: in the α case the lateral components $\mp N\sin\nu$ of the two panels are equal and opposite; in the β case the vertical components are. A *single* panel does couple, $\partial Z/\partial\beta = \pm qSa\sin\nu\cos\nu$. The sign in (8) is the usual fin sign: wind from starboard pushes the tail to port.

(a) normal force of the right panel



(b) both panels, pure α



$Z = 2N\cos\nu,\ N \propto \alpha\cos\nu$
$\Rightarrow Z \propto \alpha\cos^2\nu$

Figure 3: (a) The normal force N splits into a vertical part $N\cos\nu$ and a lateral part $N\sin\nu$. (b) For pure α both panels carry the same $N \propto \alpha\cos\nu$; the lateral parts cancel, the vertical parts add to $Z \propto \alpha\cos^2\nu$. For pure β the panels carry opposite $N \propto \pm\beta\sin\nu$; the vertical parts cancel and the lateral parts add to $Y \propto -\beta\sin^2\nu$.

4 Step 4 — Which lift-curve slope, and why a single $\cos\nu$ is wrong

Which a. The panel builds its circulation in its own plane, so a belongs to the *true, developed* planform — not to its shadow on the horizontal or vertical plane. For the limit $\nu \to 0$ to reproduce a flat tail, the aspect ratio must be that of the flattened pair:

$$A = \frac{(2l)^2}{S_{\text{tot}}} = \frac{2l^2}{S}, \tag{9}$$

with l the panel length along its span (root to tip), not the semi-span value l^2/S.

Rotation invariance. The response of the panel set to a crossflow $\mathbf{w} = V(0, -\beta, \alpha)$ is linear, $\mathbf{F} = \mathsf{K}\,\mathbf{w}/V$, with

$$\mathsf{K} = q\,a\sum_i S_i\,\mathbf{n}_i\mathbf{n}_i^{\mathsf{T}}, \qquad K_{zz} = qa\sum_i S_i\cos^2\nu_i, \quad K_{yy} = qa\sum_i S_i\sin^2\nu_i. \tag{10}$$

$\mathbf{n}\mathbf{n}^{\mathsf{T}}$ does not depend on which way $\mathbf{n}$ points. Rotating the whole tail about x only rotates K, so its trace $K_{yy} + K_{zz} = qa\sum_i S_i$ cannot change. This is the identity $\cos^2\nu + \sin^2\nu = 1$.

The rejected variants.

- *Projected area* $S\cos\nu$ with $a(A)$ drops the incidence projection. At $\nu = 45°$ it gives $S_h + S_v = \sqrt{2}\,S_{\text{tot}}$: a tilted flat surface would gain 41 % effectiveness from nothing, violating the trace invariance.
- *Projected aspect ratio* $(l\cos\nu)^2/(S\cos\nu) = A\cos\nu$ describes the shadow, which carries no circulation.

Fig. 4 compares the two laws.

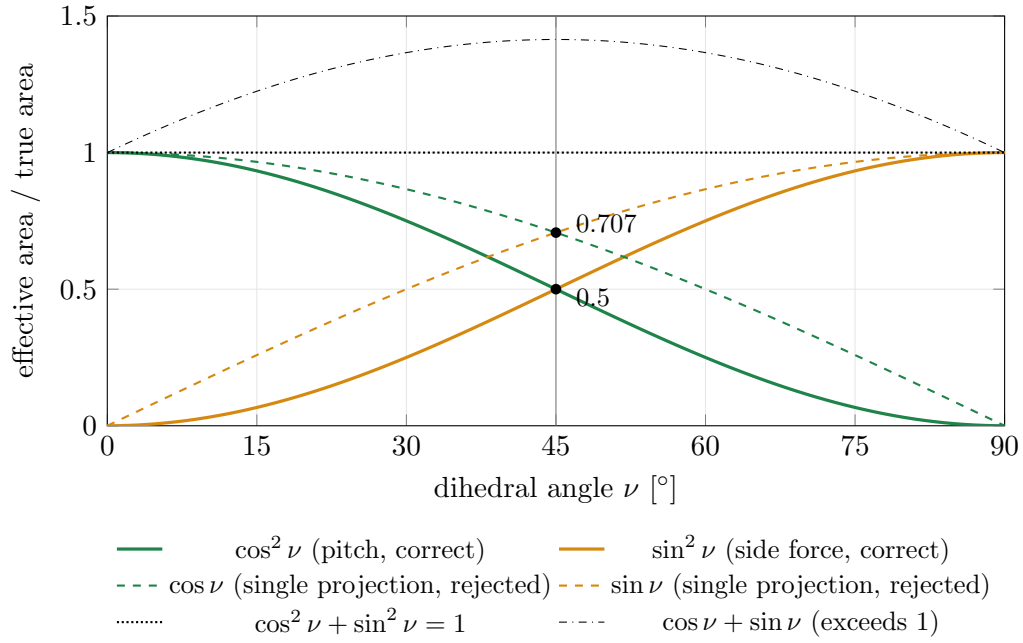


Figure 4: Effective-area factors over ν. The correct laws cross at $\nu = 45°$ with 0.5 each and always sum to 1. The single-projection variant gives 0.707 each at 45°, and its sum reaches $\sqrt{2}$.

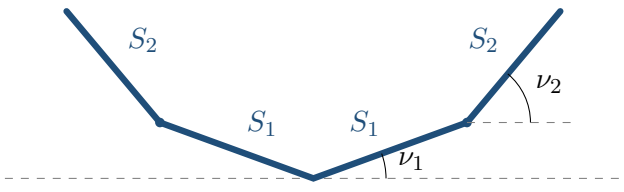
5 Step 5 — Polyhedral tails: summing segments

Assumptions. (a) Small angles and attached flow, so each segment's force is linear in its own $\alpha_{n,i}$. (b) The same normal-force slope per unit area on every segment; strictly the weight is the local load, and tip segments carry less than their area share. (c) Mutual induction between segments and between tail and fuselage is fully absorbed into the common a.

Equation. By linearity, (10) holds for any set of flat segments, giving

$$\boxed{S_{h,\text{eff}} = \sum_i S_i \cos^2\nu_i, \qquad S_{v,\text{eff}} = \sum_i S_i \sin^2\nu_i,} \qquad S_{h,\text{eff}} + S_{v,\text{eff}} = \sum_i S_i. \tag{11}$$

The off-diagonal term is $K_{yz} = -qa\sum_i S_i \sin\nu_i \cos\nu_i$ with ν_i signed (+ right, − left). It vanishes for a mirror-symmetric layout. For a layout that is not mirror-symmetric (single panel, inverted Y, an unmirrored kink) it does not vanish: α then produces side force and β produces lift, and that coupling must be carried rather than dropped.



$$S_{h,\text{eff}} = 2\,(S_1\cos^2\nu_1 + S_2\cos^2\nu_2)$$
$$S_{v,\text{eff}} = 2\,(S_1\sin^2\nu_1 + S_2\sin^2\nu_2)$$
$$S_{h,\text{eff}} + S_{v,\text{eff}} = 2\,(S_1 + S_2)$$

Figure 5: Polyhedral V-tail with an inner segment at ν_1 and an outer segment at ν_2 per side (rear view, $\nu_1 = 20°$, $\nu_2 = 50°$). Each segment contributes with its own $\cos^2\nu_i$ and $\sin^2\nu_i$.

6 Step 6 — Limits and checks

case	$S_{h,\text{eff}}$	$S_{v,\text{eff}}$
$\nu = 0$ (flat tail, aspect ratio A of (9))	S_{tot}	0
$\nu = 45°$ (equal split)	$S_{\text{tot}}/2$	$S_{\text{tot}}/2$
$\nu \to 90°$, single panel (fin of area S)	0	S
any ν (trace invariance)	$S_{h,\text{eff}} + S_{v,\text{eff}} = S_{\text{tot}}$	

For the pair, $\nu \to 90°$ is only a formal limit, because the two panels would coincide. Sizing consequence: matching a required S_h and S_v gives

$$\tan^2 \nu = \frac{S_v}{S_h}, \qquad S_{\text{tot}} = S_h + S_v. \tag{12}$$

7 Step 7 — What is exact, what is approximate, what is not captured

Exact in linear theory. Equations (4) and (5) are exact first-order statements (boundary condition and Kutta–Joukowski). For a mirror-symmetric tail the α problem loads the panels symmetrically and the β problem antisymmetrically, and superposition is exact. The true linear result is therefore

$$\frac{\partial Z}{\partial \alpha} = qS_{\text{tot}}\, a_{\text{sym}}(\nu, A, \text{fuselage}) \cos^2 \nu, \qquad \frac{\partial Y}{\partial \beta} = -qS_{\text{tot}}\, a_{\text{anti}}(\nu, A, \text{fuselage}) \sin^2 \nu. \tag{13}$$

The $\cos^2 \nu$ and $\sin^2 \nu$ factors and the cancellation of the cross terms are exact.

Approximate. Setting $a_{\text{sym}} = a_{\text{anti}} = a(A)$ of the isolated flat wing is an approximation.

- Symmetric (pitch) loading: the circulation runs continuously through the root, so the pair acts roughly like the developed wing, $a_{\text{sym}} \approx a(A)$.
- Antisymmetric (sideslip) loading: in the developed view the root circulation is zero, a vortex is shed at the root, and each panel behaves more like a surface of aspect ratio about $A/2$. The fuselage partly offsets this by acting as an end plate.
- Hence in general $a_{\text{anti}} < a_{\text{sym}}$. Panel-to-panel induction at a dihedral angle also makes both slopes depend on ν.

This interference between panels, and between panels and fuselage, is what NACA Report 823 (Purser and Campbell) measured experimentally. Neither (11) nor (13) predicts its size.

Not captured.

- Downwash and sidewash gradients; they enter only through the local α and β.
- Fuselage carry-over.
- Nonlinear effects: the panel with the larger $|\alpha_n| = |\alpha| \cos \nu + |\beta| \sin \nu$ stalls first under combined α and β.
- Moments from the side force acting at the height of the panel centroid (roll coupling).
- **Control power.** A ruddervator deflection adds $\tau\delta$ directly to α_n, so only the force projection applies:

$$\frac{\partial Z}{\partial \delta_{\text{sym}}} = qS_{\text{tot}}\, a\, \tau \cos \nu, \qquad \left| \frac{\partial Y}{\partial \delta_{\text{anti}}} \right| = qS_{\text{tot}}\, a\, \tau \sin \nu. \tag{14}$$

  These are single factors, not squared. The $\cos^2 \nu$ law must not be applied to control power.

Summary

1. **Exact (linear theory):** the $\cos^2 \nu$ and $\sin^2 \nu$ factors (one from the incidence, one from the force direction), the cancellation of the cross terms for mirror-symmetric tails, additivity over segments, and $S_{h,\text{eff}} + S_{v,\text{eff}} = S_{\text{tot}}$.

2. **Assumption:** one lift-curve slope $a(A)$ of the true developed planform ($A = 2l^2/S$ per panel) for both pitch and sideslip, with no interference beyond a. In reality $a_{\text{anti}} < a_{\text{sym}}$; area weighting of segments assumes uniform load per unit area.

3. **Check for the canon:** the limit $\nu \to 0$ must give $S_{h,\text{eff}} = S_{\text{tot}}$ and $S_{v,\text{eff}} = 0$, with $a(A)$ of the flattened pair. Companion test: $S_{h,\text{eff}} + S_{v,\text{eff}} = S_{\text{tot}}$ for every ν. Control power uses a single $\cos \nu$ or $\sin \nu$.